Fiscal Year:	FY 2017	**Task Last Updated:**	FY 02/08/2017
PI Name:	Raber, Jacob Ph.D.		
Project Title:	The Relation Between Cognitive Injury, Network Stability, and Epigenetic Change Following Exposure to Space Radiation		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Radiation health		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	raberj@ohsu.edu	**Fax:**	FY
PI Organization Type:	UNIVERSITY	**Phone:**	503-494-1524
Organization Name:	Oregon Health & Science University		
PI Address 1:	Behavioral Neuroscience and Neurology		
PI Address 2:	L470, 3181 SW Sam Jackson Park Road		
PI Web Page:			
City:	Portland	**State:**	OR
Zip Code:	97239-3011	**Congressional District:**	1
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Radiobiology NNJ12ZSA001N
Start Date:	12/21/2012	**End Date:**	12/20/2017
No. of Post Docs:	2	**No. of PhD Degrees:**	4
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	4
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Simonsen, Lisa	**Contact Phone:**	
Contact Email:	lisa.c.simonsen@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/20/2017 per NSSC information (Ed., 2/2/2017) NOTE: End date changed to 12/20/2016 per S. Monk/HRP SR (Ed., 11/4/15)		
Key Personnel Changes/Previous PI:	Febuary 2017 note: Lance Johnson obtained a tenure track faculty position at U Kentucky.		
COI Name (Institution):	Impey, Soren (Oregon Health & Science University) Turker, Mitchell (Oregon Health & Science University) Rosi, Susanna (University of California San Francisco)		
Grant/Contract No.:	NNX13AD60G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Space radiation poses a significant hazard to astronaut health both during the mission and at later times. However, uncertainty exists with respect to the level of risk to the CNS (central nervous system) following radiation doses relevant to those encountered during space missions. Radiation exposure in the brain affects the hippocampus, a structure critical for cognition. This proposal addresses section 1.2 of NASA Research Announcement NNJ12ZSA001N entitled, “Risk of Acute or Late Central Nervous System Effects from Space Radiation.” The proposed work focuses on changes in hippocampal network stability, including epigenetic mechanisms, in the development of cognitive dysfunction after space irradiation. Radiation exposure of the CNS significantly affects the hippocampus. Our preliminary data show impaired cognition in mice at two weeks following low dose space radiation. During spatial exploration, when an animal enters a specific location to which the cell responds (known as the cell's place field), hippocampal pyramidal neurons in the CA areas display robust activity. The place fields create an environment-specific "map," thought to be critical in establishing the context of a memory and resulting in the transcription of the immediate early gene (IEG) activity-regulated cytoskeletal associated protein (Arc). Thus, analyzing Arc provides a reliable method to detect the number of active cells forming environment-specific maps and to capture the environmental specificity of place fields. That is, place fields are highly reproducible and stable across multiple visits to the same environment. In contrast, this stability is not seen across visits in different environments. The proposed work will focus initially on effects of space radiation on the stability of hippocampal networks, assessed by analyzing the expression of Arc and other genes important for memory formation. We hypothesize that the detrimental effects of space radiation on hippocampus-dependent cognition are associated with changes in hippocampal network stability. Promoter region DNA methylation (5mC) of Arc leads to decreased expression and space radiation induces changes in DNA methylation. Moreover, DNA methylation plays a key role in regulating expression of genes required for spatial learning and memory. Thus, study of Arc expression and DNA methylation after charged particle exposure is warranted, followed by a more global search for DNA methylation changes in genes involved in learning and memory. A second form of DNA methylation, termed 5-hyrdroxymethylation (5hmC), is abundant in the brain and found at promoters of active genes, unlike 5mC, which is found at the promoters of inactive genes. In this proposal, we will test the hypotheses that the detrimental effects of space radiation on cognition are associated with Arc-related changes in the hippocampal networks involved in learning and memory and that these changes are related to space radiation-induced perturbation of 5mC and/or 5hmC. Our approach will link assessment of hippocampus-dependent cognitive changes, hippocampal network stability, and alterations in global DNA methylation and expression of genes involved in learning and memory. In Specific Aim 1, we will test the hypothesis that low-dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on cognitive performance in object recognition and spatial learning and memory requiring navigation. In Specific Aim 2, we will test the hypothesis that low dose whole body irradiation with 56Fe or 28Si ions or protons has short- and long-term effects on hippocampal networks involving Arc, and that cognitive injury following space irradiation is associated with changes in hippocampal network stability. In Specific Aim 3, we will test the hypothesis that cognitive changes following space irradiation are associated with alterations in DNA methylation and expression of Arc and other genes involved in learning and memory.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Humans are exposed to ionizing irradiation under a variety of conditions associated with medical treatment. While the risks of developing cognitive impairments in those situations are fairly well established, uncertainty exists with respect to other types of radiation exposure, such as those experienced in the space environment. There is a critical need to establish the level of risk to the CNS following radiation doses relevant to those encountered during space missions. NASA has identified four categories of radiation risk in the Integrated Research Plan (IRP) and their associated research and technology gaps. This project is pertinent to 1.2 “Risk of Acute or Late Central Nervous System Effects from Space Radiation” and to Space Radiation Risks CNS1, CNS2, CNS6, Degenerative 1, and Degenerative 2. The project focuses on the role of changes in hippocampal network stability and epigenetic mechanisms involved in these changes in the development of cognitive dysfunction after space irradiation. CNS-related topics expressly targeted by the IRP and that are specifically or indirectly addressed in the current project include early and late effects of space irradiation on cognitive function and cellular and molecular biomarkers that might be associated with cognitive injury. Innovative technologies are being developed for integrated functional assays to assess effects of space radiation on brain function.
	In 2016, we published a paper in BMC Genomics regarding the 20-week proton data demonstrating that proton irradiation induces persistent and tissue-specific DNA methylation changes in the left ventricle and hippocampus. The DNA methylation data showed tissue-dependent effects of proton irradiation and revealed significant major pathway changes in response to irradiation that are related to known pathophysiologic processes. Many regions affected in the ventricle mapped to genes involved in cardiovascular function pathways, whereas many regions affected in the hippocampus mapped to genes involved in neuronal functions. In the ventricle, increases in 5hmC were associated with decreases in 5mC. We also observed spatial overlap for regions where both epigenetic marks decreased in the ventricle. In hippocampus, increases in 5hmC were most significantly correlated (spatially) with regions that had increased 5mC, suggesting that deposition of hippocampal 5mC and 5hmC may be mechanistically coupled. These results demonstrate long-term changes in DNA methylation patterns following a single proton irradiation, that these changes are tissue specific, and that they map to pathways consistent with tissue specific responses to proton irradiation. Further, these results suggest novel relationships between changes in 5mC and 5hmC. In 2016, we also published a second paper in BMC Genomics. This paper was regarding the 2- and 20-week 56Fe data. BMC Genomics selects articles that are of particular interest to the community to have a complimentary video summary created by their affiliates Research Square. Our article (Short- and long-term effects of 56Fe irradiation on cognition and hippocampal DNA methylation and gene expression) was selected to have a video created which is being promoted on the journal website and social media. The youtube link is as follows: https:// The link that directs people to this publication is: (https://). In this study, we examined object recognition in 6-month-old mice irradiated with 56Fe ions, a biological age more relevant to the typical ages of astronauts. Moreover, because the mechanisms mediating the detrimental effects of 56Fe ions on hippocampal function are unclear, we examined changes in hippocampal networks involved in synaptic plasticity and memory, gene expression, and epigenetic changes in cytosine methylation (5mC) and hydroxymethylation (5hmC) that could accompany changes in gene expression. We assessed the effects of whole body 56Fe ion irradiation at early (2

Task Progress:	weeks) and late (20 weeks) time points on hippocampus-dependent memory and hippocampal network stability, and whether these effects are associated with epigenetic changes in hippocampal DNA methylation (both 5mC and 5hmC) and gene expression. At the two-week time point, object recognition and network stability were impaired following irradiation at the 0.1 and 0.4 Gy dose, but not following irradiation at the 0.2 Gy dose. No impairments in object recognition or network stability were seen at the 20-week time point at any irradiation dose used. Consistent with this pattern, the significance of pathways for gene categories for 5hmC was lower, though not eliminated, at the 20-week time point compared to the 2-week time point. Similarly, significant changes were observed for 5mC gene pathways at the 2-week time point, but no significant gene categories were observed at the 20-week time point. Only the 5hmC changes tracked with gene expression changes. Thus, dose- and time-dependent epigenomic remodeling in the hippocampus following 56Fe ion exposure correlates with behavioral changes. In 2016, we also finalized the analysis regarding the regarding the 2- and 20-week proton data. Six-month-old mice were sham irradiated or irradiated with proton (150 MeV, 1Gy). We are currently preparing a manuscript with these data. For the novel object recognition test, the mice were habituated to an open field for three subsequent days. On day 4, the mice were exposed to the open field containing two objects. On day 5, the mice were exposed to the open field with two objects but one familiar object (present on day 4) was replaced by a novel one. The time spent with each object was recorded by video and hand-scored by a researcher blinded to the treatment of the mice. Detrimental effects of proton irradiated were seen when object recognition was assessed. While sham-irradiated showed object recognition and spent significantly more time exploring the novel than the familiar object, irradiated mice did not. Spatial learning and memory was assessed in the water maze. A circular pool (diameter 140 cm) was filled with water made opaque with nontoxic chalk (24°C) and mice were trained to locate a submerged platform. To determine if irradiation affected the ability to swim or learn the water maze task, mice were first trained to locate a clearly marked platform (visible platform, Days 1 and 2). The mice were subsequently trained to locate the platform when it was hidden beneath the surface of opaque water (Days 3–5). Training during the hidden platform sessions (acquisition) required the mice to learn the location of the hidden platform based on extra-maze cues. For both visible and hidden sessions, there were two daily sessions, morning and afternoon, which were 2-h apart. Each session consisted of two trials (with 5-min inter-trial intervals). A trial ended when the mice located the platform. Mice that failed to locate the platform within 60 s were led to the platform by placing a finger in front of their swim path. The mice were taken out of the pool after they were physically on the platform for a minimum of 3 s. During visible platform training, the platform was moved to a different quadrant of the pool for each session. For the hidden platform training, the platform location was kept constant. The mice were placed into the water facing the edge of the pool in one of nine randomized locations. The start location was changed for each trial. To measure spatial memory retention, probe trials (platform removed) were conducted 24 h after the last hidden trial of each mouse on the second day of hidden platform training and 72 h following the last hidden trial of each mouse on the third day of hidden platform training and the time spent in the target quadrant as compared to any other quadrant during the probe trials was analyzed. When mice were trained to locate a visible and subsequent hidden platform in the water maze, both groups improved their performance with training but there was no effect of radiation. However, effects of proton irradiation were seen when spatial memory retention was assessed in the first probe trial (no platform) 24 hours after the second day of hidden platform training. Sham-irradiated mice showed spatial memory retention and spent more time in the target quadrant that had contained the platform during the hidden platform sessions, but proton-irradiated mice did not. In the second probe trial, 72 hours following the third day of hidden platform training sham-irradiated mice showed spatial memory retention and spent more time in the target quadrant than any other quadrant. Proton-irradiated mice still did not show spatial memory retention; they spent more time in the target quadrant than the left and right quadrants, but not than the opposite quadrant. Similar to what was seen at the 2-week time point, there were detrimental effects of proton irradiated on object recognition at the 20-week time point. The cognitive impairments were associated with reduced ability of hippocampal neurons to recognize similar environments and to discriminate distinct environments and alterations in specific pathways in hippocampus. There was a clear relationship between changed gene expression and the profiles after of DNA methylation, an epigenetic effect, following exposure, suggesting that altered gene ontology pathways account for altered cognition. Significant overlap was observed between DNA methylation changes at the 2 and 20-week time points demonstrating specificity and retention of changes in response to radiation. Moreover, a novel class of DNA methylation change was observed. These changes occurred relatively rapidly. The changes seen mapped to genes encoding neuronal functions including postsynaptic gene ontology categories. Thus, the brain's response to proton irradiation triggers a complex remodeling at some gene-associated loci in a spatially-restricted manner. Comparison of changes in DNA methylation following irradiation with protons and 56Fe showed an overlap in changes in distinct pathways. Thus, proton irradiation is both specific and prolonged and involves novel remodeling of non-random regions of the epigenome.
Bibliography Type:	Description: (Last Updated: 10/30/2023)
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